

# Assessing the Effect of Unclear Task Instructions on a Generalist Language-Conditioned Policy

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**Abstract**—Multimodal robot policies supporting text inputs, which have become an increasingly common choice for robot control, offer the potential to interface naturally with humans and receive task specifications directly from non-technical users. However, language as used in communication between humans tends to be unclear in ways that current policies may not be able to parse correctly. In this paper, we focus on two classes of unclear language: semantic ambiguity and task relevance. We introduce methods for quantifying these classes and assess the extent to which the presence of these classes in a task instruction affects policy performance on that task. To do this, we evaluate the performance of Physical Intelligence’s  $\pi_0$  model on a restricted set of tasks provided by the LIBERO benchmark. Our collected data indicate that the semantic ambiguity of an instruction, but not the task relevance, is noticeably related to its success rate. We also demonstrate the viability of an LLM-based framework for clarifying predictably constructed instructions, which could be used to mitigate ambiguity-related task failures while still allowing for communication with natural human language.

## I. INTRODUCTION

Language conditioning has emerged as an interdisciplinary field for communicating task specifications to robots through human language via a text-based modality [1]. In particular, human language is easier to humans to understand, and thus makes it easier to manually specify tasks, as well as allowing for communication of a more versatile range of abstract concepts, which modalities such as video may not be able to effectively capture. However, because human language is abstract, it may potentially carry implicit context-dependent meanings that a policy without context awareness cannot fully understand. For instance, sentences such as “Take that and put it over there with the other stuff” are commonplace in human conversation but cannot be fully understood without knowledge of the meanings of “that”, “over there”, etc. Thus, to truly accommodate natural human instructions, policies should have a sufficient level of context-awareness to infer implied specifications and successfully execute vaguely described tasks.

Specifically, we focus on two general classes of unclear language: semantically ambiguous language and task-irrelevant language. Semantic ambiguity roughly represents the degree to which a particular instruction has multiple valid interpretations that cannot be narrowed down without additional context. For example, the statement “Please fetch me some water” has an ambiguous meaning in a room with both a water bottle and a sink (which could be resolved by observing that the person issuing the statement appears thirsty or is pointing to the water

bottle). Task relevance, in contrast, represents the degree to which extra information is available in an instruction that does not contribute to specifying the immediate task. A version of the previous example with irrelevant information included is “Please fetch me my water bottle, since I just ran a marathon in 100-degree weather and I am parched”. For the purposes of determining and performing the task of fetching the water bottle, everything other than the first clause of the sentence serves purely to distract from the actual instruction.

These two examples require minimal effort to be understood by humans, and are representative of the unclear sentences which humans naturally use. Robot policies, however, tend to be trained on instructions carefully constructed to be clear, and so may struggle with interpreting unseen forms of ambiguity and irrelevance. In this project, we aim to quantifiably assess the degree to which unclear instructions affect task success rates, as a key step towards developing methods for mitigating performance degradations associated with unclear instructions.

Our project uses Physical Intelligence’s language-conditioned multimodal generalist policy  $\pi_0$  (Pi-Zero) [2] on pick-and-place tasks specified through language instructions. Experiments are set up in simulation, utilizing tasks in the LIBERO-Object task suite under the larger LIBERO dataset [3], and modified to run custom language instructions. We test instructions with varied ambiguity levels and varied relevance levels, using novel quantification metrics discussed below, finding that an increase in ambiguity induces worse task performance to some extent, while no similar effect is strongly visible for relevance. Finally, we also present an example of a closed-form instruction clarification framework, incorporating a large language model (LLM) between the user and the policy in order to reformat user instructions; in our case, we use Google’s Gemini 2.0 Flash model to test our framework. For the restricted set of tasks present in LIBERO-Object, this method is highly successful at improving overall clarity, and has potential to generalize to other task categories with predictably constructed instructions.

## II. RELATED WORK

Our study builds upon existing research in robotic manipulation, imitation learning, and language-conditioned control. Below, we summarize key contributions that inform our methodology.

**$\pi_0$ : A Vision-Language-Action Flow Model for General Robot Control**

$\pi_0$  [2] was built to serve as a general-purpose robot foundation. The generalist robot policy uses a pre-trained vision-

language model paired with a variety of manipulation tasks ranging in level of difficulty. Through the Open X Embodiment Dataset, internet-scale pretraining, and a personalized dataset provided by Physical Intelligence, this policy has been implemented by a cross-embodiment training mixture. The purpose of doing so is that it will allow the generalist robot policy to perform multiple tasks within different pre-created environments in which it has been trained in. Cross-training suggests that  $\pi_0$  has a large data set, improving overall interactions with different tasks and the success rate when asked to perform a specific task. This makes this policy effective for our project, as it gives us the flexibility to change the clarity of the instruction and clearly visualize the difference in performance.

### LIBERO: Benchmarking Knowledge Transfer for Lifelong Robot Learning

LIBERO [3] follows the concept of lifelong learning, which suggests the overall improvement of a generalist robot policy through time by learning and adapting. LIBERO contains four distinct task suites: SPATIAL, OBJECT, and GOAL, which have 10 different tasks each, and 100, which has 100 different tasks. LIBERO provides a standardized set of environments, language instructions, and tasks, which can be used to test a dataset or model; in short, it is a performance benchmark. In the case of our project, we find LIBERO really useful as we can use LIBERO to evaluate the  $\pi_0$  model, while changing the language instructions on different tasks, thus evaluating the change in the prompt’s clarity and the success rate of a task. LIBERO is also made to allow easy transfer of data, as their simulated tasks are examples of what robots should ideally be capable of performing. We can imagine how LIBERO-OBJECT could be interesting for our research as it provides an environment with multiple objects present, and the language prompt decides which items to pick, allowing us to test different classes of unclear instructions with this task suite.

### Bridging Language and Action: A Survey of Language-Conditioned Robot Manipulation

Recent advancements in language-conditioned imitation learning demonstrate how unstructured natural language can guide robotic actions. Studies such as Bridging Language and Action: A Survey of Language-Conditioned Robot Manipulation [1] explore how language can be integrated into learning-based control policies.

## III. METHOD

For this project, we introduce two novel metrics for quantifying the ambiguity and relevance of a particular task instruction with relation to the intended task:

**Ambiguity Score.** Given a particular instruction, let  $n_{total}$  be the number of pieces of information critical to accomplishing the desired task. Also let  $n$  be the number of those pieces explicitly present in the instruction, and let  $m$  be the number of those pieces implicitly present (e.g. via synonyms). Then the ambiguity score for that instruction  $\sigma$  is defined in Equation 1:

$$\sigma = \frac{n + 0.5m}{n_{total}} \quad (1)$$

**Relevance Score.** Given a particular instruction, let  $c_{total}$  be the total number of clauses in the instruction, and let  $c_{crit}$  be the minimum number of those clauses needed to unambiguously specify the desired task.<sup>1</sup> Then the relevance score for that instruction  $\eta$  is defined in Equation 2:

$$\eta = \frac{c_{crit}}{c_{total}} \quad (2)$$

While based on heuristic assessments of what information specifically is necessary to accomplish the task, these metrics form a solid foundation for specifying instruction clarity, and are adequate for the relatively simple task specifications generally given to robot policies. More detailed metrics based in formal linguistic theory, while possibly necessary for more advanced cases, are out of the scope of this project.

The basic structure of inputs for the policy is depicted in Figure 1. In this case, we use Physical Intelligence’s  $\pi_0$  model as the policy, and all observations and base instructions come from simulated tests picked from the LIBERO-Object test suite in the LIBERO dataset.<sup>2</sup> We selectively modify task instructions to either lower their ambiguity score or relevance score, and collect data on their associated success rates. We also compare these against the performance of  $\pi_0$  on two control instructions: the original task instruction, which has perfect ambiguity and relevance scores in all cases chosen for experimentation, and the sentence “Complete the task”, which has both an ambiguity score and relevance score of 0.

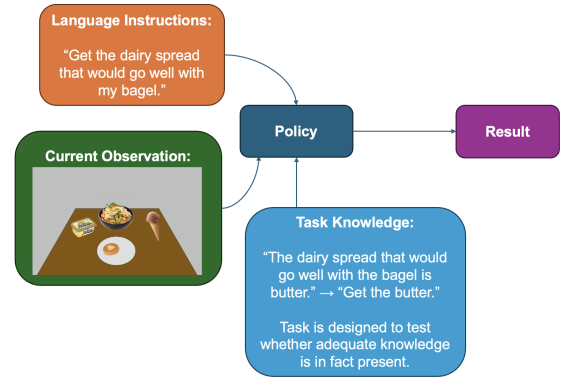


Fig. 1. Basic diagram of the inputs given to a policy for a particular task.

As a first step towards mitigating the effects of unclear human language instructions, we also explore restructuring

<sup>1</sup>Determined through the standard linguistic definition of an English clause as a grammatical unit containing a subject and a predicate.

<sup>2</sup>Originally, we planned to use the LIBERO-Spatial test suite, but found that  $\pi_0$  only rarely fully focused on custom language instructions as a result of what we believe may be overfitting to the LIBERO-Spatial test cases. Given this, and the fact that the nature of the test cases in the LIBERO-Object suite more naturally allowed for creating ambiguous instructions, we opt to use LIBERO-Object instead, though  $\pi_0$  still exhibits some problems with LIBERO-Object as discussed in the Experiments section.

TABLE I  
OBTAINED TASK SUCCESS RATES WITH AMBIGUITY TESTING

| Task # (Target Item) | Success Rate for Instruction at Given $\sigma$ |                 |                 |                 |
|----------------------|------------------------------------------------|-----------------|-----------------|-----------------|
|                      | $\sigma = 0.00$                                | $\sigma = 0.25$ | $\sigma = 0.50$ | $\sigma = 1.00$ |
| 1 (Alphabet Soup)    | 0.20                                           | 0.13            | 0.33            | 0.33            |
| 2 (Cream Cheese)     | 0.07                                           | 0.07            | 0.30            | 0.37            |
| 3 (Orange Juice)     | 0.07                                           | 0.10            | 0.30            | 0.33            |
| 4 (Ketchup)          | 0.20                                           | 0.20            | 0.33            | 0.33            |

user input via an LLM in order to automatically produce instructions with better ambiguity and relevance scores. Figure 2 illustrates this modification to the general instruction specification framework. Since the LIBERO-Object task suite is exclusively formed of pick-and-place tasks that can be expressed with the following clear instruction sentence “Pick the [item of interest] and place it in the basket”, we test Google’s Gemini 2.0 Flash model on task instructions of varying ambiguity and relevance, and attempt to restructure them in that form. A textual description of the task environment is provided in our case, which works on the restricted set of possible environments present in the LIBERO-Object task suite, but similar methodologies could be used with image or video input to multimodal LLMs in case unrestricted task environments are encountered.

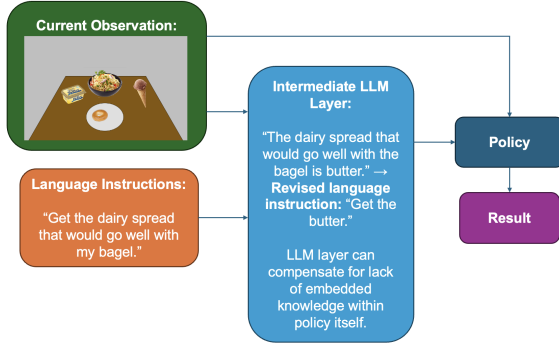


Fig. 2. Diagram of the inputs given to a policy with an intermediate LLM layer included. Note that inputs to LLM can be multimodal or textual; while we opted for the latter since textual inputs were available for our case, image or video input may be required to use in real-world situations.

#### IV. EXPERIMENTS

To test  $\pi_0$ , we select 4 different items present in the LIBERO-Object scenarios; for each item, we run  $\pi_0$  on all available simulated environments where that item is present, and instruct it to place the item in a basket with 4 different instructions with varying values of  $\sigma$  (while keeping relevance constant at  $\eta = 1$ , except when  $\sigma = \eta = 0$ ). Results are given in Table I and Figure 3. We then repeat a similar process for relevance, with results in Table II and Figure 4.

The results show that slight decreases in ambiguity and relevance scores from the maximum have a weak effect on task success rates, if any; in contrast, ambiguity scores below about 0.5 and relevance scores below about 0.25 produce significantly lower task success rates. One possible explanation

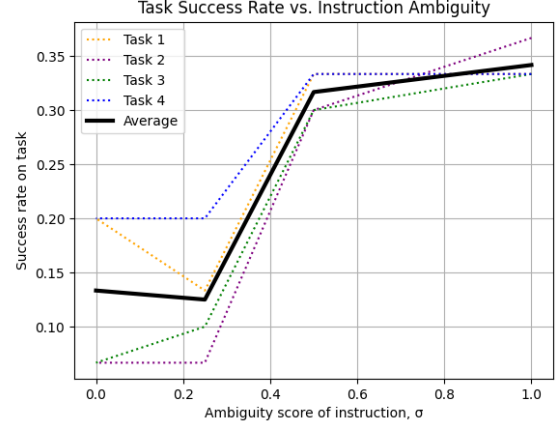


Fig. 3. Results for testing on instructions with varying ambiguity.

TABLE II  
OBTAINED TASK SUCCESS RATES WITH RELEVANCE TESTING

| Task # (Target Item) | Success Rate for Instruction at Given $\eta$ |               |               |               |
|----------------------|----------------------------------------------|---------------|---------------|---------------|
|                      | $\eta = 0.00$                                | $\eta = 0.33$ | $\eta = 0.67$ | $\eta = 1.00$ |
| 1 (Alphabet Soup)    | 0.20                                         | 0.30          | 0.30          | 0.33          |
| 2 (Cream Cheese)     | 0.07                                         | 0.23          | 0.40          | 0.37          |
| 3 (Orange Juice)     | 0.07                                         | 0.33          | 0.33          | 0.33          |
| 4 (Ketchup)          | 0.20                                         | 0.33          | 0.33          | 0.33          |

for this is that the  $\pi_0$  policy’s strategy for determining the target item in a pick-and-place task exclusively or largely involves matching the exact name of the item (as opposed to the general description of its form or functionality, as might be present in human language). This is consistent with the observation that tested sentences with higher ambiguity scores (which more often named the target items explicitly) performed better, while sentences with lower ambiguity scores (which often used synonyms or other means of implying the target items) performed worse. This explanation could also be a reason why the model was relatively resistant to changes in instruction relevance, since even with irrelevant clauses added, the target item was always clearly specified. For sentences with much lower relevance scores, however, we see non-trivial drops in model performance, which could be attributed to the increasingly likely presence of task-irrelevant nouns being present in the instruction, which the model mistakenly picks up on. Alternatively, drops in success rates coming from instructions with lower scores, which reflect a higher degree of modification to the original instruction, may simply be due to the model’s lack of training on instructions not fitting a specific template.

Note that the maximum success rate achieved on any of the instructions was no greater than 40%, even for instructions with ambiguity and relevance scores of 1.0. A possible explanation for this behavior is overfitting of  $\pi_0$  to the LIBERO-Object task suite; in the majority of LIBERO-Object test cases, the target object to be collected is placed on the right-hand side of the simulation environment, which could have taught  $\pi_0$  to look for target objects in that area even with

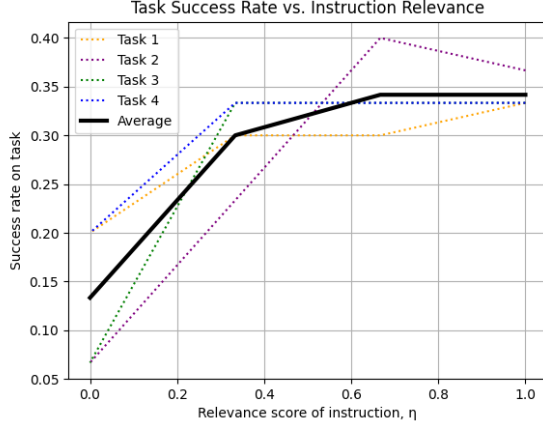


Fig. 4. Results for testing on instructions with varying relevance. Note the relatively constant success rate at most values of  $\eta$  (except near  $\eta = 0$ , where task critical information is no longer present).

custom instructions specifying new target objects in different locations. Using custom simulation environments (built in simulation frameworks such as Robosuite) were considered during the early stages of this project and could be a potential solution to this issue, as could fine-tuning of  $\pi_0$  or other models on more various task configurations. However,  $\pi_0$  seems to use provided task instructions to some extent, as can be observed in the performance changes corresponding to changes in instruction clarity, meaning it is likely that we can still reasonably assume that  $\pi_0$ ’s performance is indicative of model performance in general on unclear instructions.

We also test Gemini 2.0 Flash via the Gemini API, giving the model the task instruction as well as a textual list of possible items, one of which can be identified as the target item, then taking the identified item to form the new instruction sentence “Pick the [identified item] and place it in the basket” that could then be provided to  $\pi_0$ . The item option “none\_of\_the\_above” is also provided for instructions where the target item cannot be determined and/or is not present. We find that in all test cases, the model correctly identifies the item to collect in every single trial; the exception is when  $\sigma = \eta = 0$  and the model picks “none\_of\_the\_above” in every trial.

The reliable performance of this technique in situations with unconstrained instruction formats makes it well suited to use in the real world, with human users that commonly issue instructions with some ambiguous and/or irrelevant elements. However, while the space of possible instructions is relatively large, this technique is only well suited to restrictive tasks in which the final instruction can be expressed formulaically, as we are able to do for the pick-and-place tasks in this project. In situations where the desired tasks cannot be easily expressed using a small number of instruction “templates”, more sophisticated fine tuning or training methods for LLMs may be employed, in order to automatically produce optimally clear instructions for generalist robot policies.

Additionally, in the real world, a textual description of the environment may not be available, necessitating the use of techniques such as multimodal LLMs capable of viewing images or videos of the environment. Communication may be made even more natural for the user by integrating audio input as well, which would allow for translation of spoken human commands to well formed instructions.

## V. CONCLUSION

In this project, we evaluated the effect of semantic ambiguity and task relevance, two classes of task instruction clarity, on task success rates of the generalist language-conditioned policy  $\pi_0$ . We introduce a unique metric to quantify each class of clarity, and simulate pick-and-place tasks provided by the LIBERO benchmark with custom instructions modified for different levels of clarity. Gathered data show a relationship between decreasing task success rates and increases in both instruction ambiguity and irrelevance, though the threshold for significant task failure increases is notably more lenient for task relevance than for ambiguity. Testing of automatic LLM instruction clarification is successful even for highly unclear instructions, which shows promise for use in handling closed-form tasks similar to those tested in this project, as well as the option to generalize to a wider range of tasks through multiple possible methods.

Major limitations of this project include the flawed performance of  $\pi_0$  on custom instructions for LIBERO-Object tasks, even at maximum clarity, possibly due to issues with the model, the environment, or the experimental setup. There is potential for future projects to use different models or simulation frameworks, or to train or create new ones manually. Additionally, all tasks tested were simple pick-and-place tasks, meaning capabilities like multi-step planning, essential to successful completion of many real-world tasks, were not tested, and automated instruction clarification was relatively easy due to the predictable format of task instructions. Future work could extend the approach taken in this project to wider varieties of tasks, by more formally quantifying the differences between unclear human languages and conventionally formulaic instructions, and testing how those differences in clarity affect complex tasks that are difficult for robots, but not humans, to understand. The method of automated instruction clarification via LLMs could also be expanded to include multimodal input, handing off more of the reasoning about both the environment and task instruction to larger, more capable models. This will allow for smoother communication between users and robots in real-world situations, enabling a wider variety of successful interactions and possible tasks where robots can make significant contributions.

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